DISCOVERY OF EXOPLANETARY ATMOSPHERE WATER SIGNATURES

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Astronomers analyzing high-resolution transmission spectroscopy data from the James Webb Space Telescope
have confirmed the presence of atmospheric water vapor, carbon dioxide, and methane on exoplanet Kepler-186f.

The planet orbits within the habitable zone of its red dwarf star, located approximately 580 light-years away in the
constellation Cygnus. Spectrographic analysis revealed dense cloud decks and an atmospheric pressure comparable
to early Earth.

Astrophysicists at the Goddard Space Flight Center noted that the balance of atmospheric gases strongly suggests
photochemical processes driven by stellar ultraviolet irradiation. Further observational campaigns scheduled for 2027
will seek to detect secondary biosignatures and map atmospheric thermal phase curves across the planet's orbital cycle.